# CS 255 System Design Document

DriverPass
System Analysis
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## 1. System Overview

DriverPass is a web-based system intended to help students prepare for driving tests and arrange driving lessons. The design brings together account access, package information, online learning and practice tests, lesson scheduling, progress tracking, staff administration, reporting, and updates from the DMV. A blended design is used: the activity and sequence diagrams show how work moves through the system, while the class diagram identifies the information the system needs to store.

## 2. Design Assumptions

Students use the system through a web browser on a computer, tablet, or phone.

Staff members have different permissions based on their responsibilities.

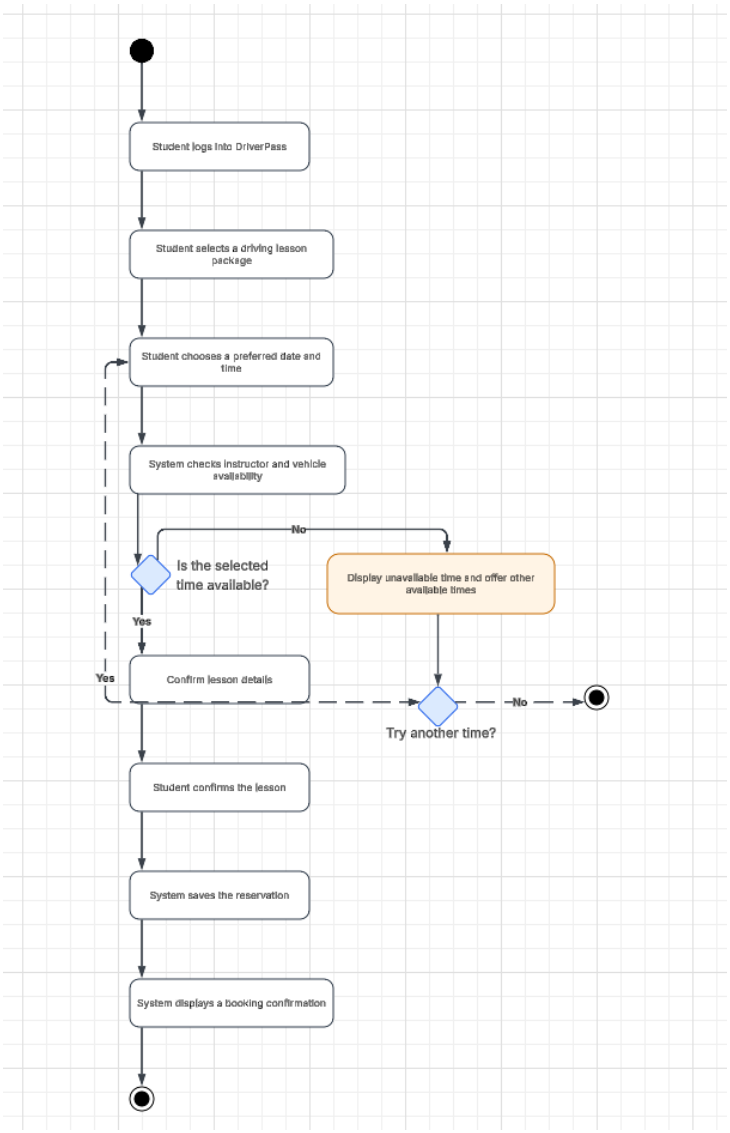
Driving lesson reservations must be checked against instructor and vehicle availability before they are confirmed.

Online lessons and practice tests record completion and results so students can review progress.

DMV information is received from an external source when an approved connection or update process is available.

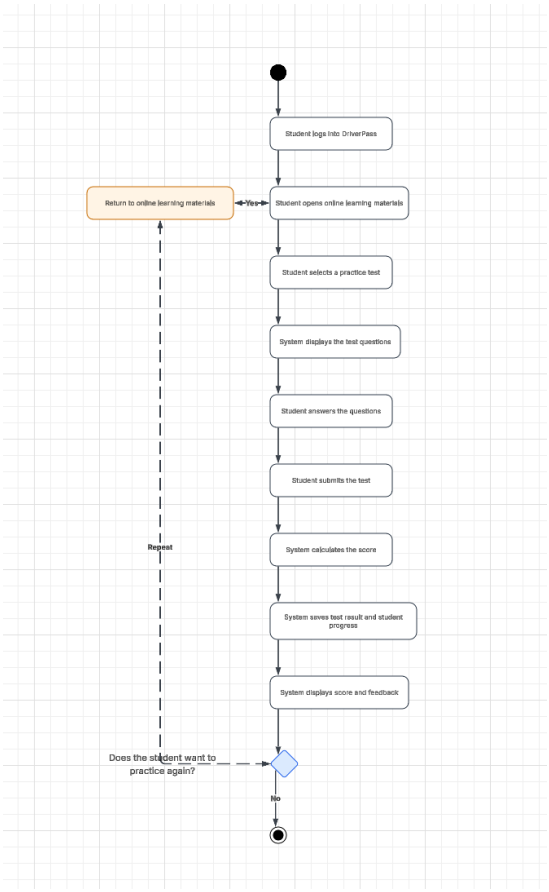
## 3. UML Use Case Diagram

The diagram summarizes the main participants and the tasks they perform. Students register, sign in, review packages, schedule lessons, and use learning materials. Secretaries maintain customer and reservation information. The owner reviews reports and business activity. The IT officer maintains system access and supports account administration. The DMV is shown as an external participant that provides relevant updates.



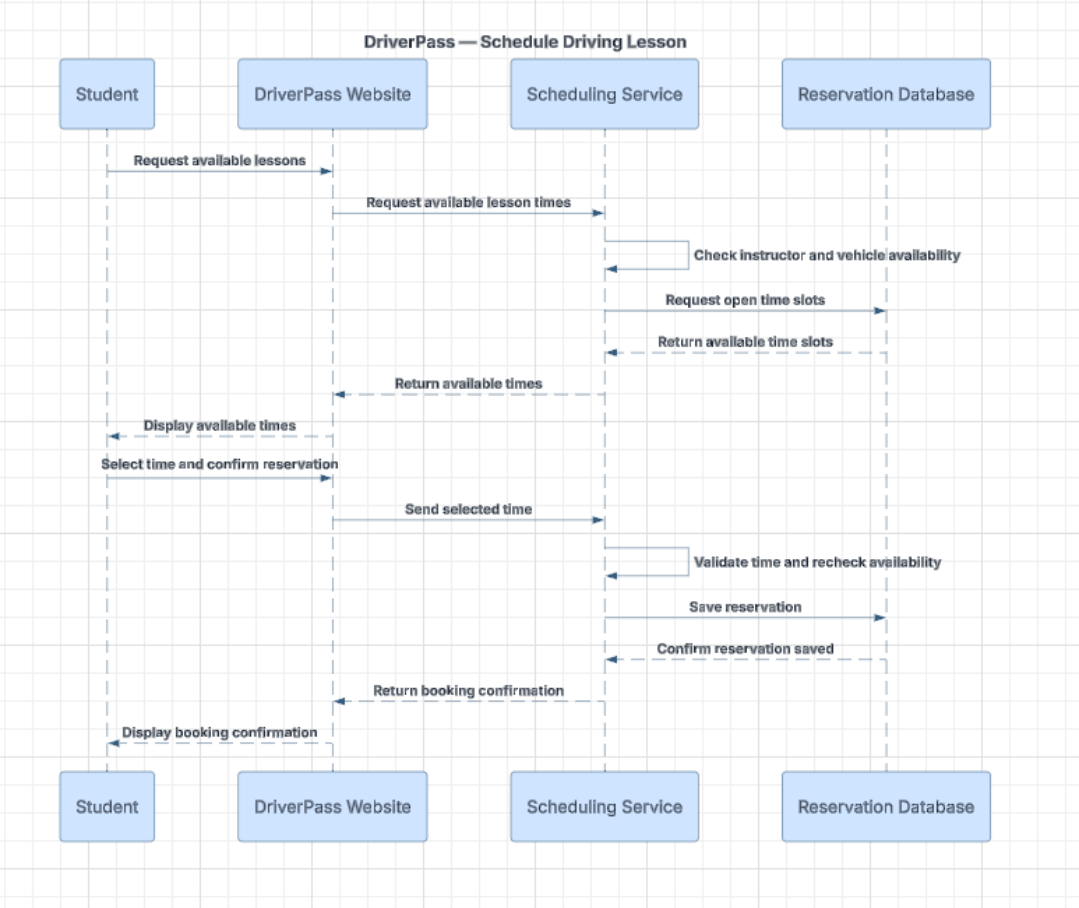

## 4. UML Activity Diagram 1: Schedule a Driving Lesson

This workflow starts when a student signs in and selects a lesson. The system checks availability before allowing confirmation. If the requested time is unavailable, the student is shown other available times. When a valid time is selected, the reservation is saved and the student is notified.



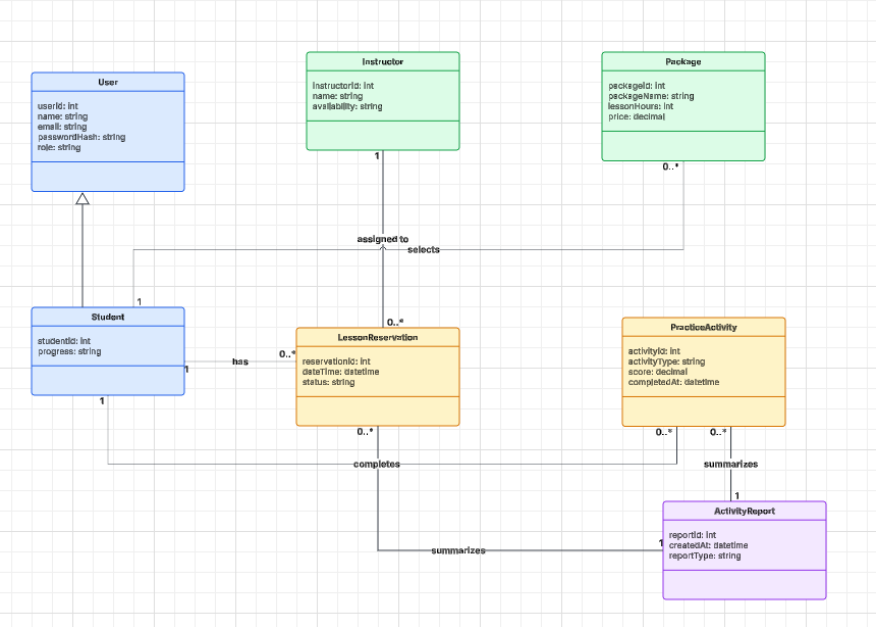

## 5. UML Activity Diagram 2: Complete Online Practice

This workflow lets a student choose an online lesson or practice test, complete the activity, and review saved progress or scores. Recording results supports the DriverPass goal of helping students prepare and identify topics that need more practice.



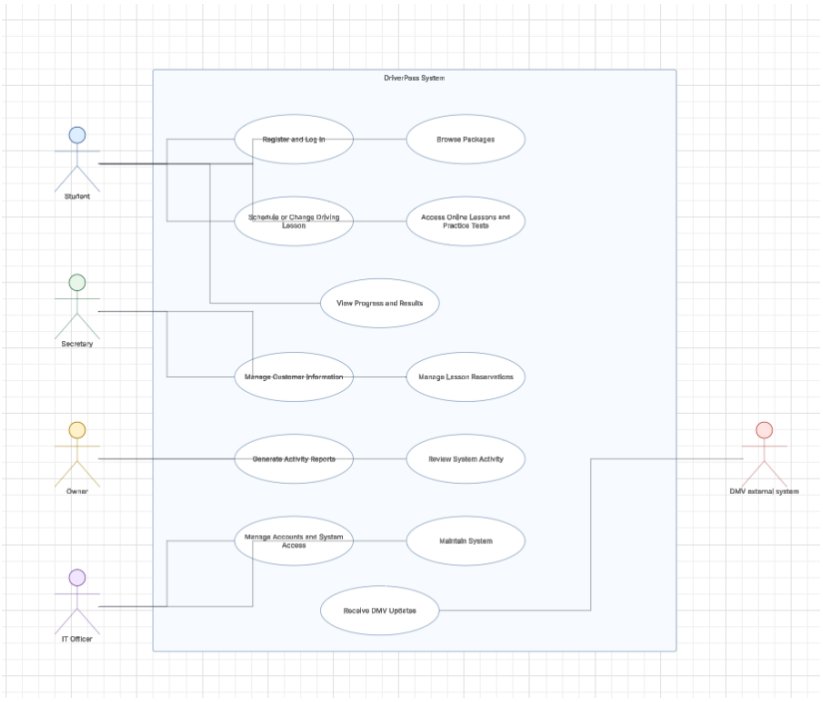

## 6. UML Sequence Diagram: Schedule a Driving Lesson

The sequence diagram shows the order of messages between the student, website, scheduling service, and reservation records. The system retrieves open time slots, presents them to the student, validates the selected slot, and saves the reservation. This separation helps prevent a lesson from being confirmed without checking availability.




## 7. UML Class Diagram

The class diagram identifies core information groups. User stores shared account details; Student represents student-specific information; LessonReservation records bookings; Instructor stores instructor information; Package defines available services and lesson hours; PracticeActivity stores online activity results; and ActivityReport supports administrative reporting. The listed attributes are a starting design and can be refined during implementation.




## 8. Technical Requirements

### Hardware

Client devices: desktop or laptop computers, tablets, and smartphones with a modern web browser and internet access.

Hosting environment: cloud-hosted application servers and database storage sized for expected student and staff usage.

Backup and recovery storage: separate protected storage for backups of system data.

### Software

A responsive web application for student and staff access.

A relational database for users, packages, reservations, instructor availability, practice results, and reports.

An authentication and authorization component that supports secure sign-in and role-based access.

An application service for scheduling, availability checks, notifications, progress tracking, and reporting.

Supported modern browsers; operating-system-specific desktop software should not be required for ordinary users.

**Tools and Infrastructure**

Lucidchart or an equivalent UML CASE tool for maintaining system diagrams.

Version control and a controlled deployment process for application changes.

Encrypted network connections (HTTPS/TLS), secure password hashing, access logging, and monitoring.

Automated backups, restore testing, and documented recovery procedures.

A controlled interface or documented staff process for receiving DMV updates; the exact integration depends on access and format provided by the DMV.

**9. Functional Requirements**

**FR-1 Accounts:** Students can register and sign in. Authorized staff can manage customer accounts and access.

**FR-2 Packages:** Students can view available packages and the services included.

**FR-3 Scheduling:** Students can schedule, change, or cancel a driving lesson, subject to the system's availability rules.

**FR-4 Learning:** Students can access online learning materials and practice tests.

**FR-5 Progress:** The system records practice completion and results so students can review progress.

**FR-6 Administration:** Authorized staff can maintain customer and reservation information and generate activity reports.

**FR-7 DMV updates:** The system supports receiving and recording relevant DMV updates when the external connection or approved update process is available.

**10. Nonfunctional Requirements**

**Security:** Use encrypted connections, securely stored password hashes, role-based permissions, and audit logs for sensitive administrative actions.

**Availability:** Host the system on reliable infrastructure and maintain backups and recovery procedures to reduce disruption.

**Usability:** Use clear navigation and responsive pages so students and staff can use the system on common devices.

**Performance:** Design common tasks such as signing in, viewing packages, and searching lesson times to respond promptly under expected demand.

**Maintainability:** Keep scheduling, account, learning, and reporting responsibilities organized so changes can be tested and deployed safely.

**Data integrity:** Validate reservations and check availability before saving a booking to reduce conflicting appointments.

**11. Security Design**

Require individual accounts rather than shared staff credentials.

Enforce permissions according to role: students access their own learning and reservation information, while staff access only the functions needed for their jobs.

Protect sign-in credentials using secure password hashing; do not store readable passwords.

Encrypt traffic between users and the system using HTTPS/TLS.

Record important account, access, and reservation changes to support review of unusual activity.

Back up data securely and limit access to backup files.

Provide a process for resetting passwords and disabling accounts when access is no longer appropriate.

**12. System Limitations and Risks**

The system depends on reliable internet access and a supported browser.

Real-time DMV updates depend on the availability, permissions, and technical format of an external DMV connection; without it, updates may need to be entered through an approved staff process.

Lesson scheduling is only as accurate as the instructor and vehicle availability information maintained in the system.

Online lessons and practice tests support preparation but do not replace supervised driving instruction or official DMV requirements.

System availability can be affected by hosting outages, maintenance, or connectivity problems.

Capacity, response time, and storage needs should be reviewed as the number of users and stored records grows.

**13. Conclusion**

This design supports DriverPass's core goals by combining student learning and lesson scheduling with staff administration, reporting, and controlled access. The diagrams provide a

shared view of system behavior and data, while the technical requirements identify the software, infrastructure, and safeguards needed for implementation.